

Lab 5

● Graded

Group

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Total Points

38 / 38 pts

Question 1

Exercise 1

6 / 6 pts

✓ - 0 pts Correct

- 0.5 pts Conclusion is insufficient or missing

- 1 pt Lower: 7.101 Upper: 7.194

Question 2

Exercise 2

8 / 8 pts

✓ - 0 pts Correct

- 1 pt Incorrect test

- 1 pt Missing or Incorrect p-value

- 1 pt Missing/Incorrect conclusion or conclusion not consistent with p-value

Question 3

Exercise 3

6 / 6 pts

✓ - 0 pts Correct

Question 4

Exercise 4

8 / 8 pts

✓ - 0 pts Correct

- 1 pt Missing/Incorrect p-value

- 1 pt Missing/Incorrect conclusion

Question 5

Exercise 5

10 / 10 pts

✓ - 0 pts Correct

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Lab 5: Hypothesis Testing & Confidence Intervals – Means

Your job: Complete Exercises 1 thru 5. This should be written in a Word document, including all appropriate explanations and labeled graphs. You are required to work with a partner. Please save your final copy as a PDF to make it easier on the grader.

North Carolina births

We will work with a random sample of observations from this [ncbirths](#) data set.

We have observations on 13 different variables, some categorical and some numerical. The meaning of each variable is as follows.

fage father's age in years.

mage mother's age in years.

mature maturity status of mother.

weeks length of pregnancy in weeks.

premie whether the birth was classified as premature (premie) or full-term.

visits number of hospital visits during pregnancy.

marital whether mother is **married** or **not married** at birth.

gained weight gained by mother during pregnancy in pounds.

weight weight of the baby at birth in pounds.

lowbirthweight whether baby was classified as low birthweight (**low**) or not (**not low**).

gender gender of the baby, **female** or **male**.

habit status of the mother as a **nonsmoker** or a **smoker**.

whitemom whether mom is **white** or **not white**.

Exercise 1: Create a 95% confidence interval for the attributes of weight and gained. For each one, report your confidence interval as a complete sentence using appropriate units. For example, we are 95% confident that the average weight of a newborn is between ___ and ___ pounds.

We are 95% confident that the average weight of a newborn is between 7.007 pounds and 7.195 pounds.

We are 95% confident that the average pregnancy weight gain for mothers is between 29.430 pounds and 31.221 pounds.

Exercise 2: Now let's explore a subset of the data. Create a subset of the data containing only premies (there should be 152 premies in your subset). You learned how to create a subset of the data in Lab 3. Test the hypothesis that the average weight of a premie is 4.9 lbs. Use $\alpha = 0.05$. Write up the full hypothesis test, including:

- null and alternative hypotheses,
- sample average, sample standard deviation, and sample size
- standard error,
- test statistic and degrees of freedom,
- p-value, and
- conclusions (relating to the context)

Null Hypothesis (H_0): The true average weight of a premie is 4.9 pounds.

Alternative Hypothesis (H_a): The true average weight of a premie is not 4.9 pounds.

Sample average: 5.128 pounds

Sample standard deviation: 1.9696 pounds

Sample size: 152

Standard error: $1.9696 / \sqrt{152} = 0.1597$ pounds

Test statistic: $t = 1.4298$

Degrees of freedom: 151

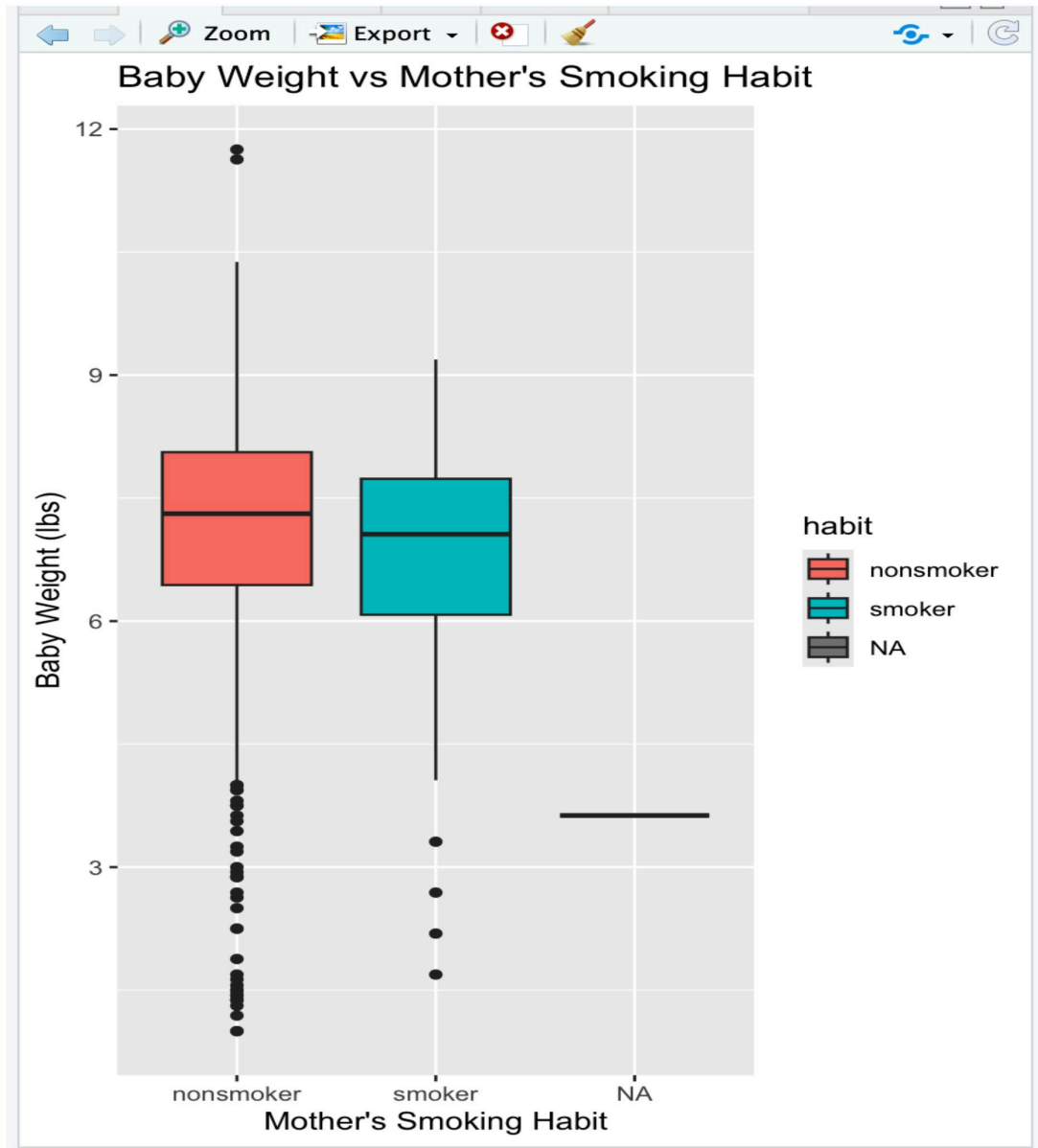
p-value: 0.1548

Conclusion:

Since the p-value is greater than the significance level of 0.05, we fail to reject the null hypothesis. There is not enough evidence to conclude that the true average weight of a premie is different from 4.9 pounds.

The 95 percent confidence interval for the true mean weight of premies is between 4.813 pounds and 5.444 pounds. Since 4.9 falls within this interval, it supports the conclusion that the average premie weight could reasonably be 4.9 pounds.

Exercise 3: Consider the possible relationship between a mother's smoking habit and the weight of her baby. Plot the data with side-by-side boxplots for smoker and nonsmoker. Provide appropriate title/labels for your graph. Then provide a table with the mean, standard deviation, and sample size for the weights of babies for the smoking and the non-smoking subsets.



Mothers Habit	Mean Weight (lbs)	Std dev (lbs)	Sample size
Nonsmokers	7.14	1.52	873
Smokers	6.83	1.39	126

Exercise 4: Conduct a hypothesis test to determine if the average weights of babies born to smoking and non-smoking mothers are different. Include:

- null and alternative hypotheses,
- standard error,
- test statistic and degrees of freedom,
- p-value, and
- conclusions (related to the context, and understandable to the general public)
- Follow up with a 95% confidence interval about the difference in average weights if appropriate.

Null Hypothesis (H_0): There is no difference in the average weights of babies born to smoking and non-smoking mothers. (mean smoker = mean nonsmoker)

Alternative Hypothesis (H_a): There is a difference in the average weights of babies born to smoking and non-smoking mothers. (mean smoker $\neq$ mean nonsmoker)

Standard error: 0.1338 pounds

Test statistic: $t = -2.359$

Degrees of freedom: 171.32

p-value: 0.01945

Conclusion:

Since the p-value is less than 0.05, we reject the null hypothesis. There is statistically significant evidence that the average weights of babies born to smoking and non-smoking mothers are different. Babies born to mothers who smoke tend to weigh less, on average, compared to babies born to non-smoking mothers.

Follow-up 95 percent confidence interval:

The 95 percent confidence interval for the difference in average weights is from -0.580 pounds to -0.052 pounds.

Since the entire confidence interval is negative, it supports the conclusion that babies born to smoking mothers tend to weigh less than babies born to non-smoking mothers.

Exercise 5: Conduct a hypothesis test to determine if the average weights of babies born to married and non-married mothers are different. Include:

- side by side boxplots
- null and alternative hypotheses,
- sample averages, sample standard deviations, and sample sizes
- standard error,
- test statistic and degrees of freedom,
- p-value, and
- conclusions (related to the context, and understandable to the general public)
- Follow up with a 95% confidence interval about the difference in average weights if appropriate.

Null Hypothesis (H_0): There is no difference in the average weights of babies born to married and non-married mothers. (mean married = mean not married)

Alternative Hypothesis (H_a): There is a difference in the average weights of babies born to married and non-married mothers. (mean married $\neq$ mean not married)

Sample averages:

- Married mothers: 7.18 pounds
- Not married mothers: 7.03 pounds

Sample standard deviations:

- Married mothers: 1.48 pounds
- Not married mothers: 1.56 pounds

Sample sizes:

- Married mothers: 664
- Not married mothers: 336

Standard error: 0.1027 pounds

Test statistic: $t = 1.429$

Degrees of freedom: 687.47

p-value: 0.1534

Conclusion:

Since the p-value is greater than 0.05, we fail to reject the null hypothesis. There is not enough

evidence to conclude that there is a difference in the average weights of babies born to married and non-married mothers.

In simpler words, whether the mother was married or not does not appear to significantly affect the baby's birth weight.

Follow-up 95 percent confidence interval:

The 95 percent confidence interval for the difference in average weights is from -0.056 pounds to 0.354 pounds.

Since the interval includes 0, it supports the conclusion that there is no significant difference between the two groups.

